Manufacturing & Video-Analytics Intelligent Report

“I want the data from the mes application for machine utilization and from video analytics I want camera configuration information”

Generated: 2026-10-18 17:49:40
Time filter applied

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Executive Summary

2	12	333	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, configuration, machine, utilization

Telemetry window: 2026-10-17T12:19:40Z → 2026-10-18T12:19:40Z (bucketed every 5m)

Video Analytics Camera Configurations & Status

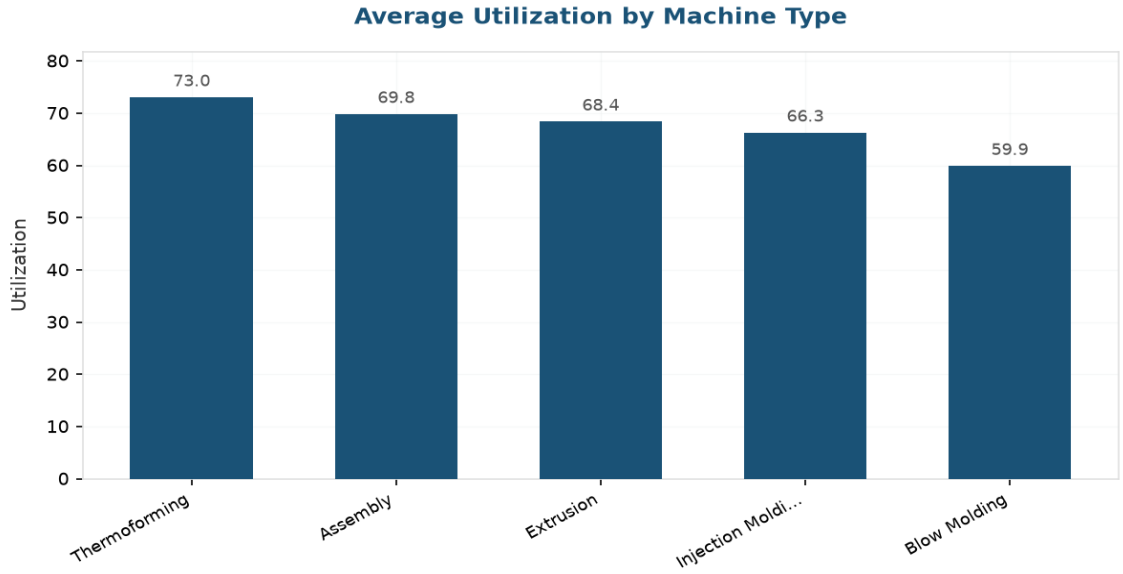
The Video Analytics system manages configuration details for 3 registered cameras: 'Flarehub' (camera 1), 'Luxsphere' (camera 2), and 'Entry Gate' (camera 6), with 'Entry Gate' specifically mapped to Plant 1, Department 1, and Location 1. All 3 cameras currently report an active 'online' status under user 'admin'. However, the system's status log table records 50 historical status entries across camera IDs 1 through 3, all logging 'offline' events. This indicates historical connectivity drops or network instability that should be audited alongside current live feeds.

Data preview — cameras (showing 3 of 3 points/rows)

camera_number	department_id	location_id	name	plant_id	status	user_id
1	None	None	Flarehub	None	online	admin
2	None	None	Luxsphere	None	online	admin
6	1	1	Entry Gate	1	online	admin

MES Machine Utilization by Equipment Type

The MES MachineMaster registry tracks 40 operational machines divided across 5 main types: Injection Molding, Thermoforming, Assembly, Extrusion, and Blow Molding. Fleet utilization averages 68.61% overall, ranging from a low of 43.40% to a high of 92.70%. Total capacity per machine averages 541.73 units with an average hourly capacity of 216.60 units/hr. The spread in utilization highlights significant operational variation between highly utilized production lines and under-utilized assets.

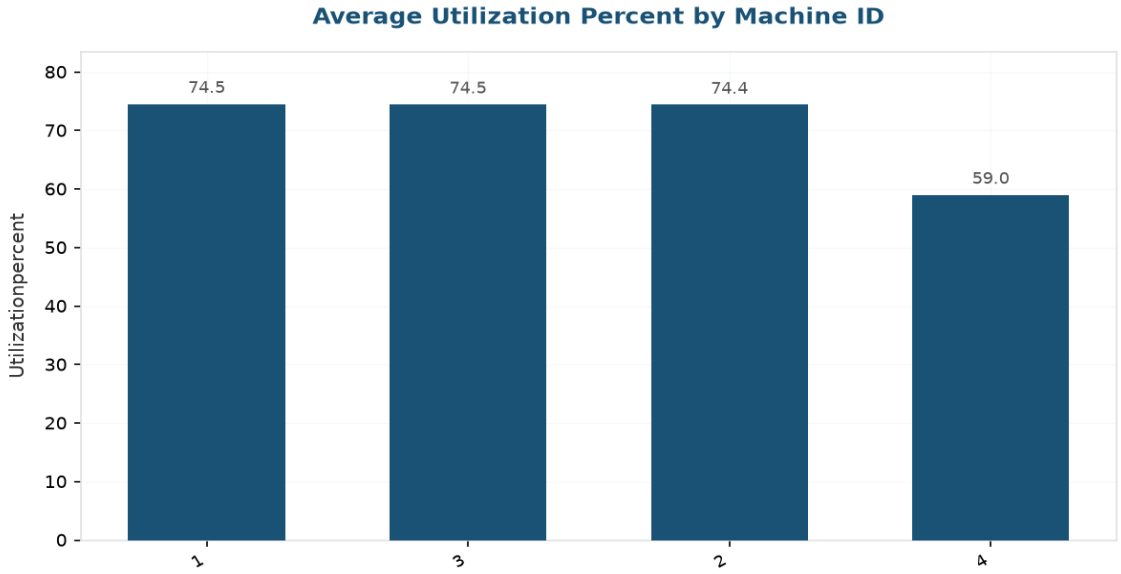


Data preview — MachineMaster (showing 10 of 40 points/rows)

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
81.90	M001	1	Injection Molding Line 1	Injection Molding	772.96	178.64	1
45.00	M002	2	Thermoforming Line 2	Thermoforming	554.94	77.34	1
58.90	M003	3	Thermoforming Line 3	Thermoforming	826.22	229.64	1
75.90	M004	4	Thermoforming Line 4	Thermoforming	681.29	150.48	1
59.00	M005	5	Assembly Line 5	Assembly	120.46	235.09	1
69.70	M006	6	Injection Molding Line 6	Injection Molding	190.81	333.32	1
43.40	M007	7	Extrusion Line 7	Extrusion	705.28	335.53	1
56.50	M008	8	Thermoforming Line 8	Thermoforming	572.02	66.41	1
77.10	M009	9	Assembly Line 9	Assembly	283.00	388.73	1
70.40	M010	10	Blow Molding Line 10	Blow Molding	943.44	217.79	1

MES Shift-Level Capacity Analysis

The CapacityAnalysis dataset records 50 shift-level utilization measurements across machines 1 through 4 and shifts 1 through 4. Utilization across individual shifts averages 72.92%, spanning from a minimum shift performance of 37.50% to a maximum peak of 100.00%. The wide range indicates shift scheduling imbalances or batch downtime events on key equipment during specific shift windows.



Data preview — CapacityAnalysis (showing 10 of 50 points/rows)

MachineId	UtilizationPercent	CapacityId	CustomerId	ShiftId
1	100.00	1	1	4
1	51.62	2	1	1
1	52.62	3	1	4
1	100.00	4	1	3
1	100.00	5	1	2
1	61.88	6	1	2
1	67.25	7	1	1
1	59.75	8	1	3
1	64.00	9	1	4
1	76.38	10	1	1

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	71.4%	0	Needs review
camera_status_logs	50	100.0%	0	Good
MachineMaster	40	100.0%	0	Good
CapacityAnalysis	50	100.0%	0	Good
Machine	40	100.0%	0	Good
MachineFGMapping	50	100.0%	0	Good
MachineMaterialMapping	50	100.0%	0	Good
WeighingMachine	50	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,372	2,557	7,929	$0.0311